

are done on most of the image data, such as instance visualisation, frequency analysis or noise reduction. On the other hand, there are some specific operations like masking Fourier transform to eliminate some frequency noise on certain images.

It is not always practical to have a C++ code for every minor task. The user, who may use a particular piece of code, probably just once in his lifetime, should not be spending time in editing and compiling it. This might not be economical.

User interfaces

For datasets varying from single-dimensional scalar signals to multi-dimensional multi-spectral volumetric images, various user interfaces (UIs) present in this toolkit provide mechanisms for conversion, manipulation, filtering and visualisation of images.

Command-line interface. G'MIC language is a powerful script language for image-processing pipelines. The command line interface (CLI), gmic, allows users to access features of the framework from within a shell. In the absence of a dedicated graphical interface, CLI is the most powerful interface that allows users to work on sequences of float-valued 3D volumetric images having arbitrary channels, without limits other than the available memory.

By default, loading and saving of image sequences is done using OpenCV. Users can enter demo mode, displaying all capabilities of the tool by invoking gmic on the command line without any other arguments.

Gmic_gimp plugin. GIMP is a very popular image retouching and editing software. gmic_gimp plugin renders the capabilities of G'MIC to GIMP. More than 400 filters including film emulation, frames, layers, light and shadows, patterns and rendering are available with the tool.

ZArt Interface. ZArt interface was initially developed as a demon-



Fig. 3: G'MIC Colorization algorithm for colourisation of an old black-and-white photograph



Fig. 4: An implementation of G'MIC colour-transfer algorithm

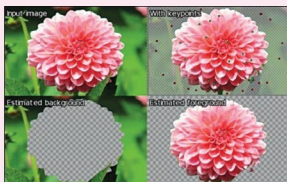


Fig. 5: Decomposition of foreground and background of an image

stration platform, where a user can process images from a webcam. Now this interface is capable of importing any kind of image files. Most filters from GIMP plugin have now been shifted to ZArt interface.

G'MIC online. Image-processing algorithms from a Web browser can be directly applied on the images with the help of this Web service.

Besides this, there is a thread-safe C++ library, libgmic, to be linked to third-party applications. This multi-threaded library implements the entire G'MIC core image-processing routines and structures.

New features in G'MIC

Let us take a look at some of the latest features introduced in the software.

For colour processing.

Interactive colour curves filter.

Instead of going for conventional red-green-blue (RGB) colour space,

G'MIC relies on an interactive colour curves filter that includes CMY, CMYK, HSI, HSL, HSV, Lab, Lch and YCbCr. This can be done using the new command line instruction `-x_color_curves`. Plugin users, who use Colors / Curves [interactive] filter can serve their favourite colour curve and use it repeatedly.

Comics colourisation. Even using a computer, the activity of comic colourisation is a tedious task. The new G'MIC command `-x_colorize` and corresponding Black & White / Colorize [interactive] filter in G'MIC plugin helps the user to do the colourisation job easily.

The user simply has to put some keypoints here and there within different image regions. G'MIC provides a minimal interface for doing. By analysing contours and colour-point interpolation, the algorithm comes up with a probable colourisation.

Black-and-white picture colourisation. With a small change in the above algorithm, G'MIC provides a new tool to colourise old black-and-white photographs. Even for black-and-white picture colourisation, the user could effectively use Black & White / Colorize [interactive] filter.

Colour transferring. In the colour-transferring algorithm, colour of the source image is replaced by the colour schemes of a reference image, so as to get a better colour ambience for the original image. This is achieved with the help of Colors/Transfer colors [advanced] plugin.

Foreground/background extraction algorithm. In the photo